

Archaeologist

WEIGHT 3 · LONGEST · lineage first

Slogan

Label recoveries by disturbing the expert, not by deploying the learner.

Genealogy one-liner (must appear):

BC (ALVINN) → covariate shift (Ross & Bagnell) → DAgger family → DART (CoRL'17: optimized supervisor-side action noise) → [CLEAR] ACT cites then rejects DART noise for fine teleop in favor of chunking → [TENTATIVE rhyme] DP/ π /DROID diversity/chunk/FM noise — not forks.

49→79%

HSR success

3×

Humanoid wall-clock

−5% vs −80%

collect reward

+62% rel

vs BC

α inv-U

3 > 6 humans

Talk arc — why the Archaeologist seat

Lineage depth over architecture tourism

1 Priors table

ALVINN → Ross&Bagnell → DAgger → SEARN/AggreVaTe → PoE → query-eff → GAIL

CLEAR

2 DART pivot

Supervisor-side optimized action noise · Eq.3+4 · α -shrinkage

CLEAR

3 Fig.1 cartoon

Off-policy vs On-policy vs DART funnel

CLEAR

4 Full lineage

CLEAR vs TENTATIVE badges on one diagram

CLEAR

5 ACT CLEAR cite

RSS'23 [36]: noise→corrective but breaks fine teleop → chunking

CLEAR

6 TENTATIVE rhymes

DP · DROID · OXE · Octo/OpenVLA · $\pi_0/\pi_{0.5}$ FM $\neq$ DART teleop

TENTATIVE

7 Motion deep dive

Compounding T · recovery boundary · action $\neq$ state · +chunking

CLEAR

8 Close

What later learned · exam · tattoo

CLEAR

Priors — what DART builds on

Label every row CLEAR or TENTATIVE on-slide

ALVINN / BC (Pomerleau '89)

Classic off-policy supervised IL; road-edge drift folklore

CLEAR

Ross & Bagnell 2010

Formalizes covariate shift / compounding under own dist

CLEAR

Dagger (Ross et al. 2011)

On-policy iterative learner→supervisor queries — main foil

CLEAR

SEARN / AggreVaTe family

Reduction-style IL ancestors; Deeply AggreVaTeD cited [20]

CLEAR*

Persistence of excitation

Classical white-noise injection so models stay valid off-nominal

CLEAR

Query-eff / safety Dagger

Zhang&Cho; Laskey human-centric — document Dagger pain

CLEAR

GAIL supervisor setup [6]

TRPO det-mean + 50-subsample MuJoCo IL protocol

CLEAR

Noise-injection teleop folklore

Practice DART turns into optimized $\Sigma^{\alpha} + \alpha$ -shrinkage

TENTATIVE*

*SEARN: CLEAR as field prior; TENTATIVE as direct DART emphasis. *Folklore: TENTATIVE named prior; CLEAR that DART formalizes it.

DART pivot — supervisor-side optimized action noise

Eq. 3 MLE noise fit · Eq. 4 α -shrinkage · Algorithm 1

PIVOT BOX

Keep off-policy collection. Corrupt expert actions with learned noise ψ so demos already contain recovery trajectories.

Eq.3: $\hat{\psi}_{k+1} = \operatorname{argmin}_{\psi} E[-\sum \log \pi^*(\hat{\pi}(x_t) | x_t, \psi)]$ under noisy-supervisor trajs

Eq.4: $\psi^{\alpha} = \hat{\psi} \cdot \beta^*$ s.t. expected simulated error $\approx \alpha$ (Gaussian: $\Sigma^{\alpha} = \alpha / (T \operatorname{tr}(\Sigma)) \cdot \Sigma$)

Closed forms: Gaussian outer-product $\hat{\Sigma}$ · ϵ -greedy disagreement rate. Shape from learner error; magnitude from α .

NOT DAgger

Learner rollouts + supervisor labels on bad states

IS DART

Supervisor + action noise labels recoveries near manifold

IS NOT state noise

Control-stream $u \rightarrow$ dynamics induce wider state coverage

α prior

MuJoCo: $\alpha = T \operatorname{tr}(\hat{\Sigma})$
HSR: $\alpha \in \{3, 6\}$ · 3 wins

Fig. 1 — Off-policy vs On-policy vs DART

Paper cartoon (~page-04): widened supervisor funnel + recoveries

We measure the difference between controls using a surrogate loss $l : \mathcal{U} \times \mathcal{U} \rightarrow \mathbb{R}$ [14, 13]. A commonly considered surrogate loss is the squared L2-norm on the control vectors $l(\mathbf{u}_1, \mathbf{u}_2) = \|\mathbf{u}_1 - \mathbf{u}_2\|_2^2$ [8]. We measure total loss along a trajectory with respect to two policies π_{θ_1} and π_{θ_2} by $J(\theta_1, \theta_2 | \xi) = \sum_{t=0}^{T-1} l(\pi_{\theta_1}(\mathbf{x}_t), \pi_{\theta_2}(\mathbf{x}_t))$.

The objective of Imitation Learning is to minimize the expected surrogate loss along the distribution induced by the robot's policy:

$$\min_{\theta} E_{p(\xi|\pi_{\theta})} J(\theta, \theta^* | \xi). \quad (1)$$

In Eq. 1, the distribution on trajectories and the cumulative surrogate loss are coupled, which makes this a challenging optimization problem. The field of Imitation Learning has considered two types of algorithmic solutions to this objective, off-policy learning and on-policy learning [13]. A common off-policy technique is Behavior Cloning, which samples from the supervisor's distribution and performs expected risk minimization on the demonstrations:

$$\theta^R = \operatorname{argmin}_{\theta} E_{p(\xi|\pi_{\theta^*})} J(\theta, \theta^* | \xi).$$

The performance of the policy π_{θ^R} can be written in terms of the following decomposition:

$$\begin{aligned} E_{p(\xi|\pi_{\theta^R})} J(\theta^R, \theta^* | \xi) \\ = \underbrace{E_{p(\xi|\pi_{\theta^R})} J(\theta^R, \theta^* | \xi) - E_{p(\xi|\pi_{\theta^*})} J(\theta^R, \theta^* | \xi)}_{\text{Shift}} + \underbrace{E_{p(\xi|\pi_{\theta^*})} J(\theta^R, \theta^* | \xi)}_{\text{Loss}}, \end{aligned}$$

which corresponds to the covariate shift and the standard loss. In this work, we focus on minimizing the covariate shift. For references on how to minimize the standard loss see [16]. In the next section, we will show how noise injection can be used to reduce covariate shift.

4 Off-Policy Imitation Learning with Noise Injection

Noise injected into the supervisor's policy simulates error occurring during testing. Under this noise, the supervisor is forced to take corrective actions in order to successfully perform the task. The corrective examples allow the robot to learn how to recover when it deviates from the supervisor's distribution. However, because the demonstrations are still concentrated around the supervisor's policy, it can have less cognitive load on the supervisor than on-policy methods, which are concentrated around the robot's policy. The intuition of providing corrective examples during data collection is shown in Fig. 1.

Denote by $p(\xi|\pi_{\theta^*}, \psi)$ a distribution over trajectories with noise injected into the supervisor's distribution $\pi_{\theta^*}(\mathbf{u}|\mathbf{x}, \psi)$. The parameter ψ represents the sufficient statistics that define the noise distribution. For example, if Gaussian noise is injected parameterized by ψ , $\pi_{\theta^*}(\mathbf{u}|\mathbf{x}, \psi) = \mathcal{N}(\pi_{\theta^*}(\mathbf{x}), \Sigma)$.

Similar to Behavior Cloning, we can sample demonstrations from the noise-injected supervisor and minimize the expected loss via standard supervised learning techniques. This can be written as follows:

$$\theta^R = \operatorname{argmin}_{\theta} E_{p(\xi|\pi_{\theta^*}, \psi)} J(\theta, \theta^* | \xi) \quad (2)$$

Eq. 2, though, does not explicitly minimize the covariate shift for arbitrary choice of ψ . In order to reduce covariate shift, we introduce the following Lemma to bound the shift. We can then attempt



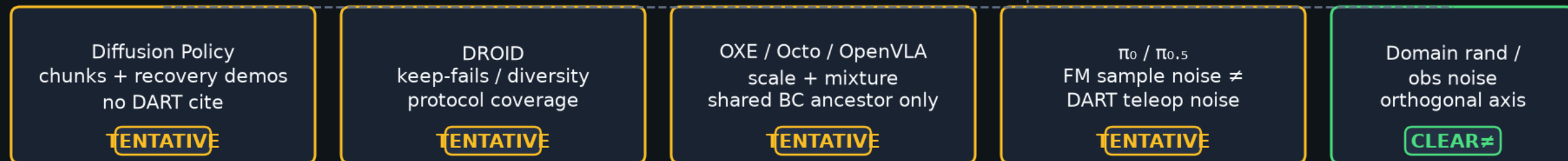
Figure 1: Robot Learning to reach a goal state $\mathbf{x}_G$. The grey denotes the distribution over trajectories. Left: Off-Policy learning in which the supervisor, the orange arrows, provides demonstrations. The robot, the teal arrows, deviates from the distributions and incurs high error. Middle: On-Policy which samples from the current robot's policy, the light teal arrows, to receive corrective examples from the supervisor. Right: DART, which injects noise to widen the supervisor's distribution and provides corrective examples. DART is off-policy but robust.

Full lineage — CLEAR vs TENTATIVE

Syllabus genealogy with badges on every node



TENTATIVE design rhymes — NOT citation forks



One-liner (say it aloud):

BC → covariate shift → DAgger → DART (optimized supervisor-side action noise) → [CLEAR] ACT cites then rejects DART noise for fine teleop in favor of chunking → [TENTATIVE] DP/π/DROID diversity/chunk/FM — not forks.

Rule: cite ACT-DART as CLEAR. Call everything else rhyme unless a local pack greps a DART cite.

ACT CLEAR cite — RSS'23 [36]

Zhao et al. ALOHA/ACT: noise yields corrective behavior but can break fine teleop

CLEAR CITATION

ACT cites DART [36] explicitly: injecting noise at demo time yields corrective / recovery behavior.

BUT: “for fine manipulation, such noise injection can directly lead to task failure.”

→ Motivates action chunking as the preferred anti-compounding tool for fine teleop.

What ACT keeps

Recovery intuition from DART is real — demos that leave the nominal manifold and return are valuable labels.

What ACT rejects

Open-loop Gaussian action noise during precision demos can ruin the demonstration itself (contact / insert / thread).

Syllabus moral

DART = data-side recovery labels
Chunking = effective-horizon fix
Complementary, not substitutes.
Task-gate the noise.

TENTATIVE rhymes — modern IL/VLA robustness

Design echoes $\neq$ citation graph. Say TENTATIVE out loud.

Diffusion Policy

TENTATIVE

Chunks + receding horizon; Push-T recovery demos. No DART cite in local txt.

DROID

TENTATIVE

Scene churn / keep-fails for recovery research. Protocol coverage, not Σ^α .

OXE / RT-X

TENTATIVE

Scale + mixture; failure-recovery segment upsampling discussions.

Octo / OpenVLA

TENTATIVE

Generalist BC/VLA; robustness from scale & co-train, not supervisor noise.

π_0 / $\pi_{0.5}$

TENTATIVE

Continuous chunks + flow matching. BEHAVIOR'25 correlated FM noise = sample-time $\neq$ DART teleop.

Domain rand / obs noise

CLEAR $\neq$

Orthogonal perception axis. CLEAR distinction — not descendants.

Motion / covariate-shift deep dive

Primary syllabus node for compounding error in IL

1 • Compounding over T

Same long-horizon pathology that later motivates ACT chunking, DP receding-horizon, RTC soft-inpaint. DART attacks the *data distribution*; chunking attacks *effective horizon*. Complementary.

2 • Recovery at the boundary

Explicit goal: label states at the edge of the expert policy so the learner can restabilize after small deviations — teleop “funnel”.

3 • Action noise $\neq$ state/obs noise

DART = control-stream disturbances $\rightarrow$ dynamics-induced state coverage. Distinct from obs dropout, domain rand, visual augment.

4 • Noise schedule matters

Iterative Σ_k^α (shape from learner error cov; scale from α). Wrong scale (isotropic I, or α too large for humans) fails. α inverted-U: HSR 3>6.

5 • Vs DAgger sampling locus

Same shift objective, opposite locus: expert neighborhood vs learner trajectory. DART keeps demos high-reward (Humanoid -5% vs -80%).

What later papers learned (+ / -)

Transferables and caution flags from the DART → 2026 arc

KEPT (+)

- + Recovery labels are a first-class data product
- + Optimized noise $\gg$ isotropic folklore
- + Off-policy collection can match on-policy robustness
- + α / magnitude is a real hyperparameter (inverted-U)
- + Safer collect-time reward is an ops metric

DROPPED / CAUTION (-)

- Fine contact teleop hates open noise (ACT)
- Humans degrade under aggressive α
- Global Σ ignores heteroscedastic $\Sigma(x)$
- 2017 CNN+open-loop grasp $\neq$ modern VLA stack
- Don't overclaim VLA "descendants" without cites

Exam bite / syllabus placement

Where DART sits in the Data week reading list

EXAM BITE

DART reframes covariate shift as a *data-augmentation* problem: inject optimized supervisor-side action noise so recovery is labeled in-distribution — safer/cheaper than DAgger, complementary to later chunking (ACT CLEAR-cites then rejects open noise for fine teleop).

Pair with

Ross & Bagnell (shift)
+ ACT or DP (chunking)
+ one 2025 BEHAVIOR writeup

Not a substitute for

On-policy query theory
VLA architecture papers
Classical motion planners

Role weight

Archaeologist = 3
Deepen prior→subsequent
CLEAR vs TENTATIVE rigor

Tattoo card — numbers every deck shares

Memorize / flash on every role seat

49%

BC HSR success

79%

DART $\alpha=3$ HSR

72%

DART $\alpha=6$ HSR

+62% rel

avg lift vs BC

3x

Humanoid wall-clock
vs DAgger

-5%

DART collect reward
vs clean supervisor

-80%

DAgger collect reward
vs supervisor

α inv-U

humans: 3 > 6

Also tattoo:

Action noise $\neq$ state/obs noise · optimized $\Sigma^\alpha \gg$ isotropic · ACT CLEAR-cites DART then prefers chunking ·
BEHAVIOR: steal recovery-demo + α -schedule — don't overclaim VLA lineage

Evidence — Fig. 2 MuJoCo curves

Reward / wall-time / collection reward · DART ≈ DAgger, safer/cheaper

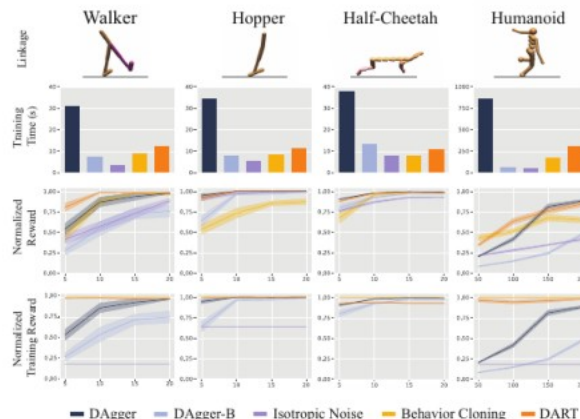


Figure 2: Top: The four different locomotive domains in MuJoCo we evaluated DART on: Walker, Hopper, Half-Cheetah and Humanoid. Top Middle: The time, in seconds, to achieve the performance level reported below. DART achieves similar performance to DAgger in all 4 domains, but requires significantly less computation because it doesn't require retraining the current robot policy after each demonstration. DAgger-B reduces the computation required by less frequently training the robot, but suffers significantly in performance in domains like Humanoid. Bottom Middle: The learning curve with respect to reward obtained during training with each algorithm plotted across the number of demonstrations. Bottom: The reward obtained during collection of demonstrations for learning. DART receives near the supervisor's reward at all iterations whereas DAgger can be substantially worse in the beginning.

policies in Tensorflow. At each iteration we collected one demonstration. In order to make the task challenging for Imitation Learning, we used the same technique as in [6], which is to sub-sample 50 state and control pairs from the demonstrated trajectories, making the learners receive less data at each iteration.

We compare the following five techniques: Behavior Cloning, where data is collected from the supervisor's distribution without noise injected; DAgger, which is an on-policy method that stochastically samples from the robot's distribution and then fully retrain the robot's policy after every demonstration is collected; DAgger-B, which is DAgger but the retraining is done intermittently to reduce computation time; Isotropic-Gaussian which injects non-optimized isotropic noise; and DART. See supplement material for how the hyper-parameters for each method are set.

We used the following MuJoCo environments: Walker, Hopper, Humanoid, and Half-Cheetah. The size of the state and control space for each task is $\{|X| = 17, |U| = 6\}$, $\{|X| = 11, |U| = 3\}$, $\{|X| = 376, |U| = 17\}$, $\{|X| = 117, |U| = 6\}$, respectively. All experiments were run on a MacBook Pro with an 2.5 GHz Intel Core i7 CPU. We measured the cumulative reward of each learned policy by rolling it for 500 timesteps, the total computation time for each method and the cumulative reward obtained during learning.

Fig. 2 shows the results. In all domains, DART achieves parity with DAgger, whereas Behavior Cloning and DAgger-B are below this performance level in Walker and Humanoid. For Humanoid, DART is 3x faster in computation time and during training only decreases the supervisor's cumulative reward by 5%, whereas DAgger executes policies that have over 80% less cumulative reward than the supervisor. The reason for this is that DAgger requires constantly updating the current robot policy and forces the robot into sub-optimal states. While, one way to reduce this computation is to decrease

Evidence — Fig. 3 HSR clutter

BC 49% · DART($\alpha=3$) 79% · DART($\alpha=6$) 72% · 4 humans · 20 evals



Figure 3: Left: Experimental setup for the grasping in clutter task. A Toyota HSR robot uses a head-mounted RGBD camera and its arm to push obstacle objects out of the way to reach the goal object, a mustard bottle. The robot’s policy for pushing objects away uses a CNN trained on images taken from the robot’s Primesense camera, an example image from the robot’s view point is shown in the orange box. Right: the Success Rate for Behavior Cloning, DART($\alpha = 3$) and DART($\alpha = 6$). DART($\alpha = 3$) achieves the largest success rate.

the number of times the policy is updated. DAgger-B illustrates that this can significantly deteriorate performance. Lastly, naively applying isotropic noise does not perform well, and leads to unsafe policies during execution, which suggests the need for optimizing the level of noise.

5.2 Robotic Grasping in Clutter

We evaluate DART with human supervisors in a grasping in clutter task on a Toyota HSR robot. We consider a task inspired by the Amazon Picking Challenge [2]. The goal of the task is for the robot to retrieve a goal object from a cupboard. The task is challenging because the goal object is occluded by obstacle objects and the robot must reason about how to clear a path based on observations taken from an eye-in-hand perspective. The objects are 6 common household food items, which consist of boxes and bottles with varying textures and mass distributions. The target object is fixed to always be a mustard bottle. The robot, task and image viewpoint are shown in Fig. 3. See supplement material for additional information on the task.

We use 4 supervisors who have robotics experience but not specifically in the field of Imitation Learning and compare Behavior Cloning and DART. When performing the study, we first collect $N = 10$ demonstrations with Behavior Cloning (i.e. no noise) and then in a counter-balanced ordering collect $N = 30$ more demonstrations with each technique. Our experiment was within-subject, have every supervisor perform all three methods.

We only updated the noise parameter after the first 10 demonstrations (i.e. $K = 2$). The final robot policy was then trained on the total of 40 demonstrations. We consider two different choices of α : $\alpha = 3\text{Tr}(\Sigma_1)$ and $\alpha = 3T\text{tr}(\Sigma_1)$. These choices correspond to the intuition that with small datasets in high dimensional image space the robot will have significant shift from the current loss on the supervisor’s distribution. Through the rest of the paper, we will write our choices as $\alpha = 3$ and $\alpha = 6$ for brevity in notation.

During policy evaluation, we measured success as 1) the robot is able to identify the goal object and 2) there being a clear path between its gripper and the object. Once these conditions are identified, as detailed in the supplement material, an open-loop motion plan is generated to execute a grasp around the target object.

In Fig. 3, we report the average performance of the three techniques across DART 20 times on different sampled initial states. DART with $\alpha = 3$ performs the best out of the three techniques, with a 79% success rate over traditional Behavior Cloning’s 49% success rate. Interestingly, DART with $\alpha = 6$ performs better than Behavior Cloning, but only has a 72% success rate. This may suggest this level of noise was potentially too high for the human supervisor.

6 Conclusion

This paper considers injecting noise into the supervisor’s demonstrations to minimize covariate shift. Our algorithm, DART, provides the robot with corrective examples at the boundary of the supervisor’s policy. By being concentrated around the supervisor’s policy, it collects demonstrations without visiting highly sub-optimal states and is easier for human supervisors tele-operating. We demonstrate it achieves parity with on-policy methods in simulated MuJoCo domains and is significantly better than traditional Behavior Cloning on a real robot. For code and more information see <http://berkeleyautomation.github.io/DART/>

Appendix — covariate shift + Wishart $\text{Tr}(\Sigma)$

Fig.4 shift gap closes · Fig.5 magnitude critical ($\text{tr} \approx 0.5$ can match DART)

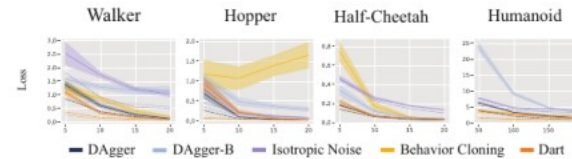


Figure 4: The covariate shift on the four different locomotive domains in MuJoCo we evaluated DART on: Walker, Hopper, Half-Cheetah and Humanoid. The dashed lines correspond to the surrogate losses on the supervisor distributions, $E_{p(\xi) \sim p_{\text{sup}}} J(\theta^H, \theta^* | \xi)$, and the solid lines correspond to the loss on the robot distributions $E_{p(\xi) \sim p_{\text{rob}}} J(\theta^H, \theta^* | \xi)$. DART is able to reduce the covariate shift in all four domains, which is illustrated by the dashed line converging to the solid line. Additionally, DAgger is also able to reduce the shift. However, Behavior Cloning is less effective at doing so, which is illustrated by a slower convergence rate.

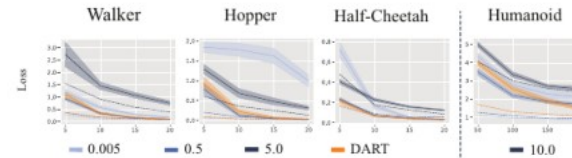


Figure 5: The losses of learners trained from supervisors with randomly sampled covariance matrices are compared against the losses of DART. Again, the dashed lines represent the loss on the supervisor distributions while the solid lines represent the loss on the robot distributions. When the simulated error is well-close, such as $\text{Tr}(\Sigma) = 0.5$ in these experiments, the performance matches that of DART and the covariate shift is reduced. However higher or lower levels of the noise can cause drastically higher losses.

open-loop motion plan is generated to execute a grasp around the target object. In order for the robot to identify the target object, we use the Faster R-CNN trained on the PASCAL VOC 2012 dataset, which has a bottle object category [12]. Once the mustard bottle has been identified and the a path is cleared a human supervisor tells the robot to execute the open-loop motion plan towards a fix position that the goal object is at. The human supervisor is used to ensure that the success criteria is not prone to error in the system, but the supervisor is blinded to which policy is being evaluated to prevent biases. We evaluate the policy 20 times on different sampled initial states. A video of the learned policy and the robot setup can be found here: <https://youtu.be/LfMD6911esg>.

Close — Archaeologist takeaways

Say the genealogy; police CLEAR vs TENTATIVE

1. Slogan: Label recoveries by disturbing the expert, not by deploying the learner.
2. Pivot: supervisor-side optimized action noise (Eq.3+4) — off-policy antidote to shift.
3. CLEAR fork: ACT cites DART then rejects open noise for fine teleop → chunking.
4. TENTATIVE only: DP/DROID/OXE/ π FM noise = rhymes, not citation children.
5. Complementary: DART = data-side recovery; chunking = horizon — use both wisely.
6. Tattoo: 49/79/72 · +62% · 3× · −5% vs −80% · α inverted-U (3>6).